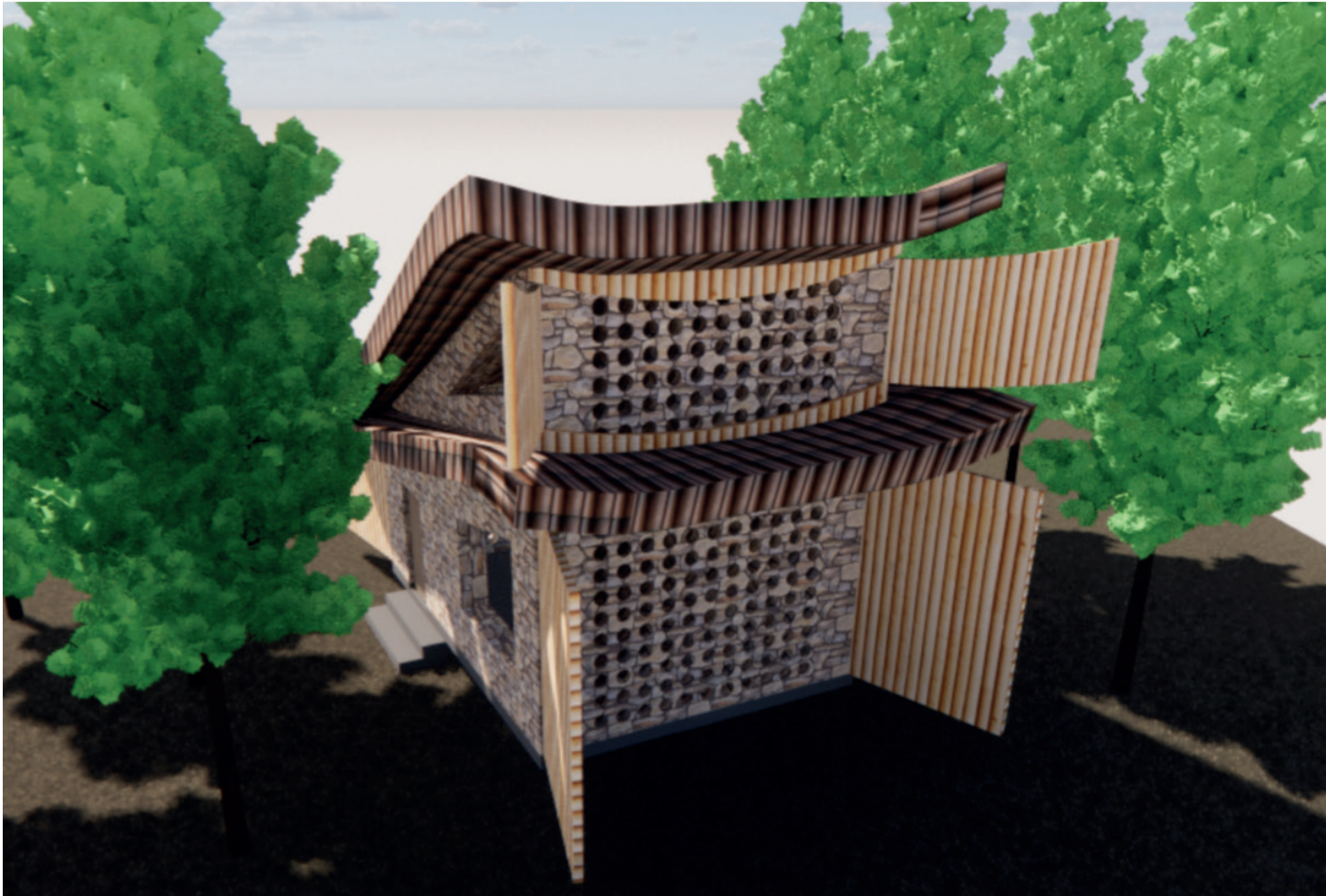
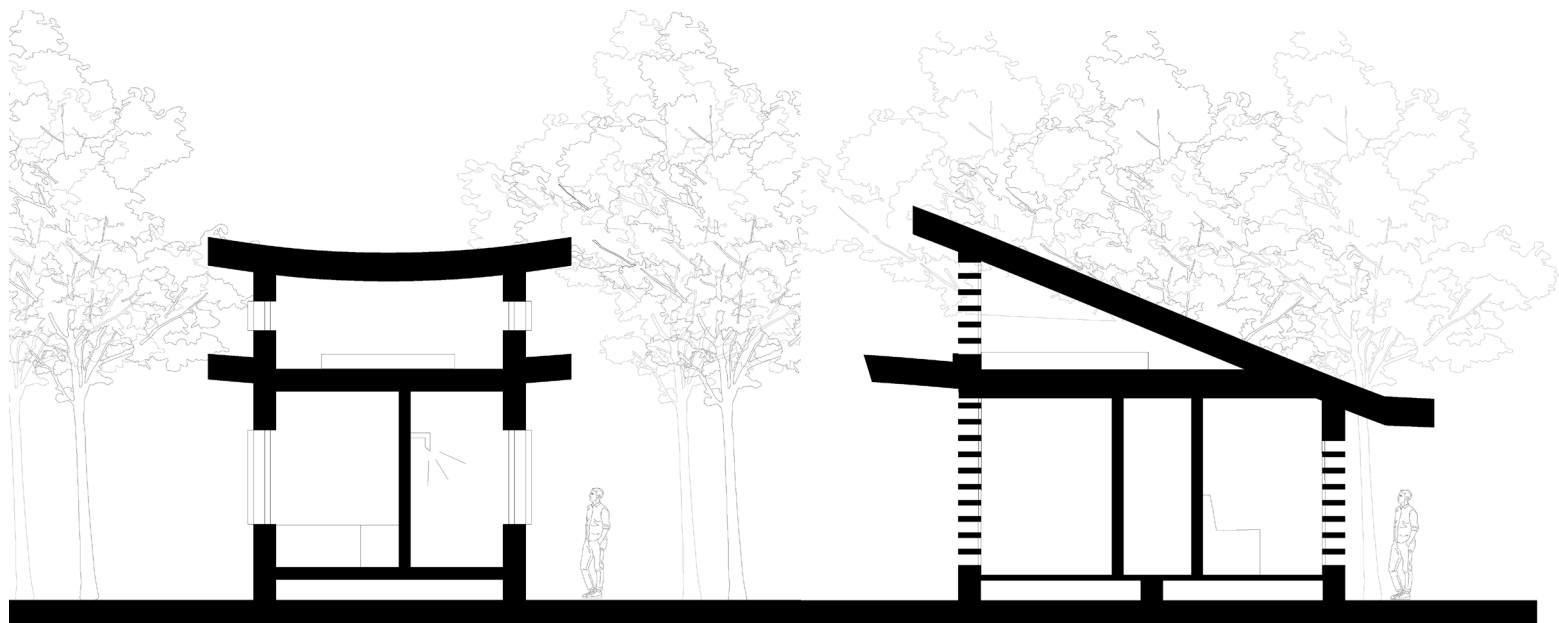


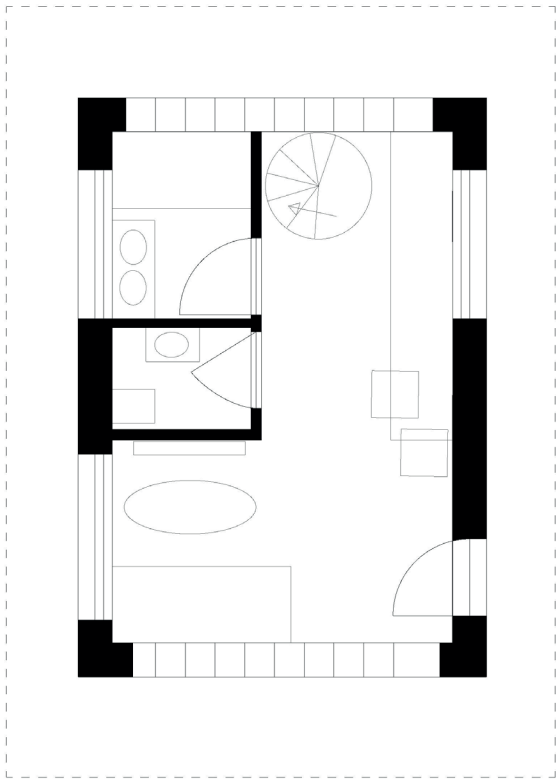
The windchime



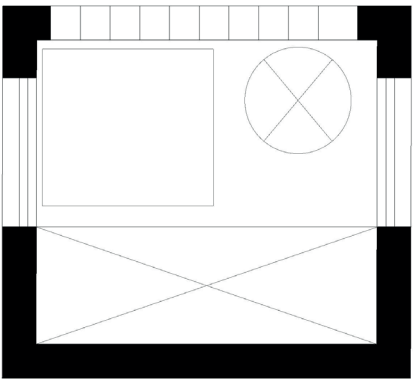
Sections



Floorplans



Groundfloor



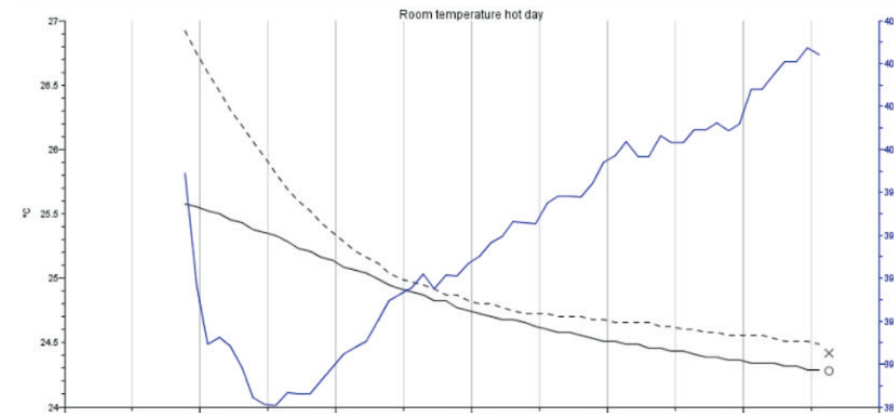
Secondfloor /loft

Het experiment

Hoeveel verschilt de temperatuur bij gebruik van watervasthoudend materiaal?

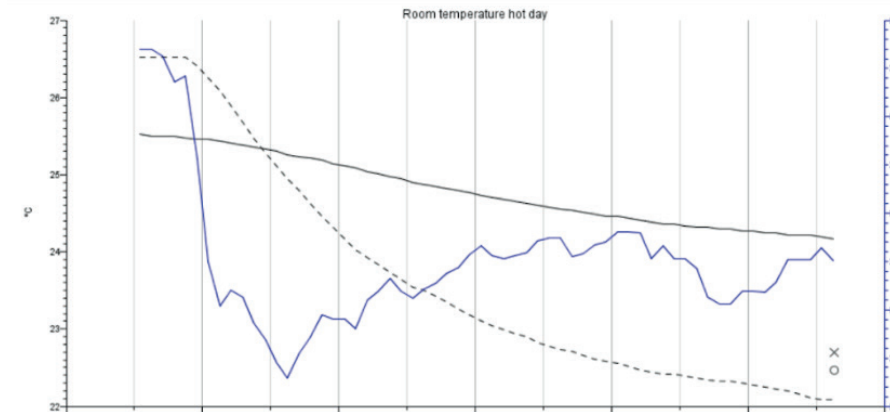
Blokken zonder wateropname:

- Grote gaten
iets meer dan 2 graden verschil na 10 min, gelijk aan buiten na half uur

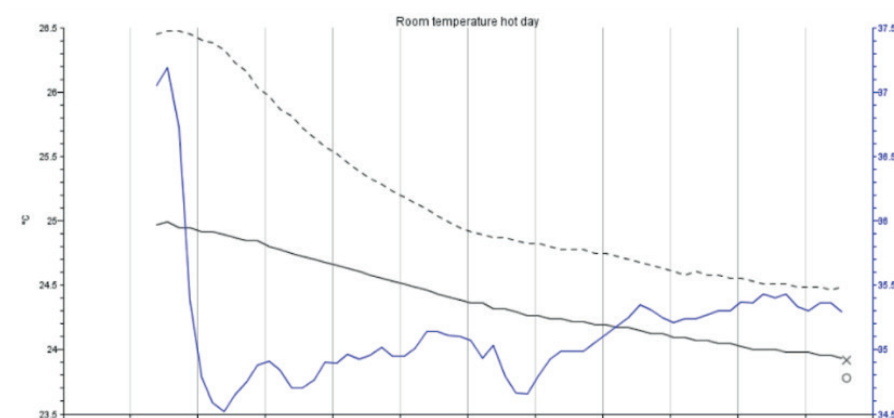


Blokken met wateropname:

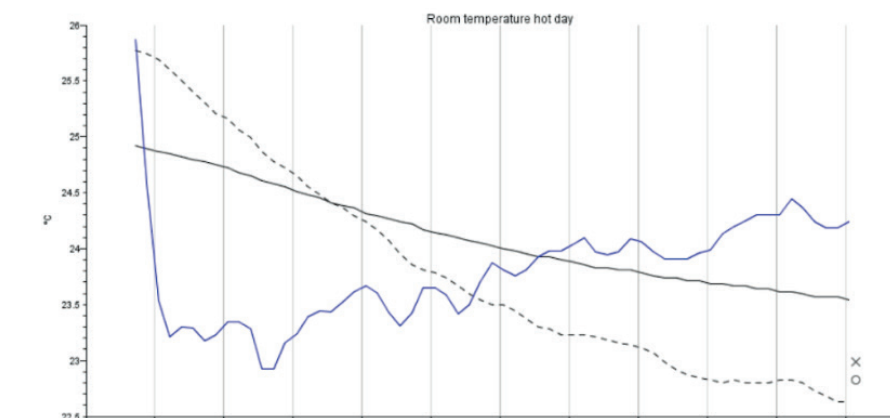
- Grote gaten
iets meer dan 3,5 graden verschil na 10 min.



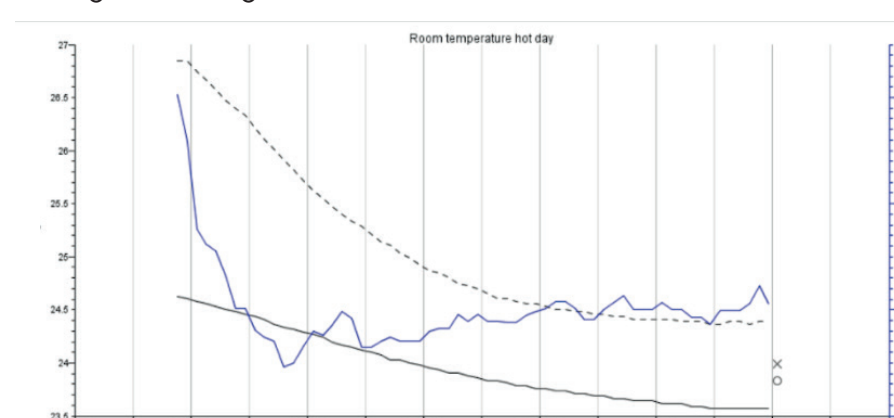
- Middel grote gaten
ongeveer 1,5 graden verschil na 10 min.



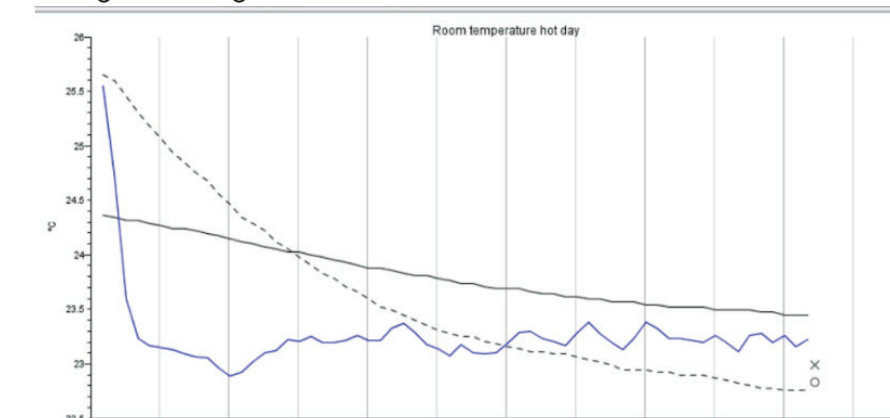
- Middel grote gaten
iets minder dan 3,5 graden verschil na 10 min.



- Kleine gaten
ongeveer 1,5 graden verschil na 10 min.

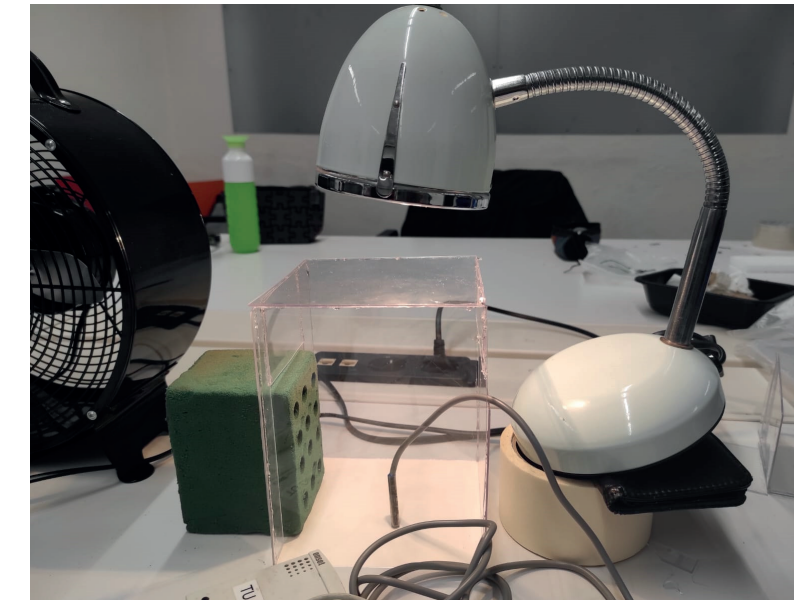


- Kleine gaten
ongeveer 3 graden verschil na 10 min.



— Binnen temperatuur
- - Buiten temperatuur
— Vochtigheid

De opstelling:



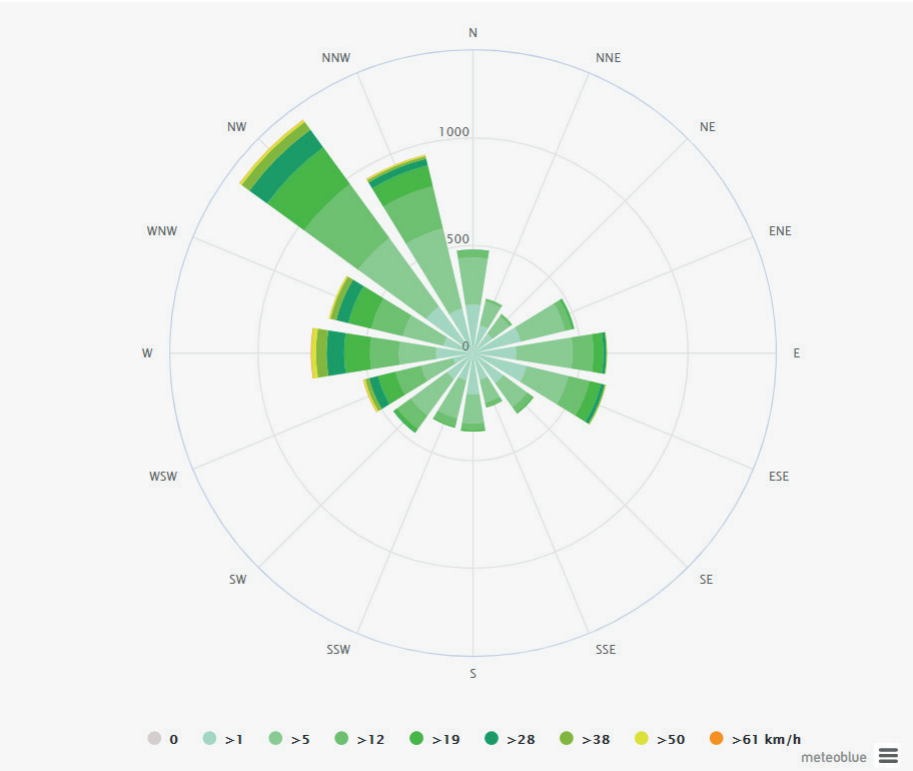
Inspiratie 'Cool bricks'



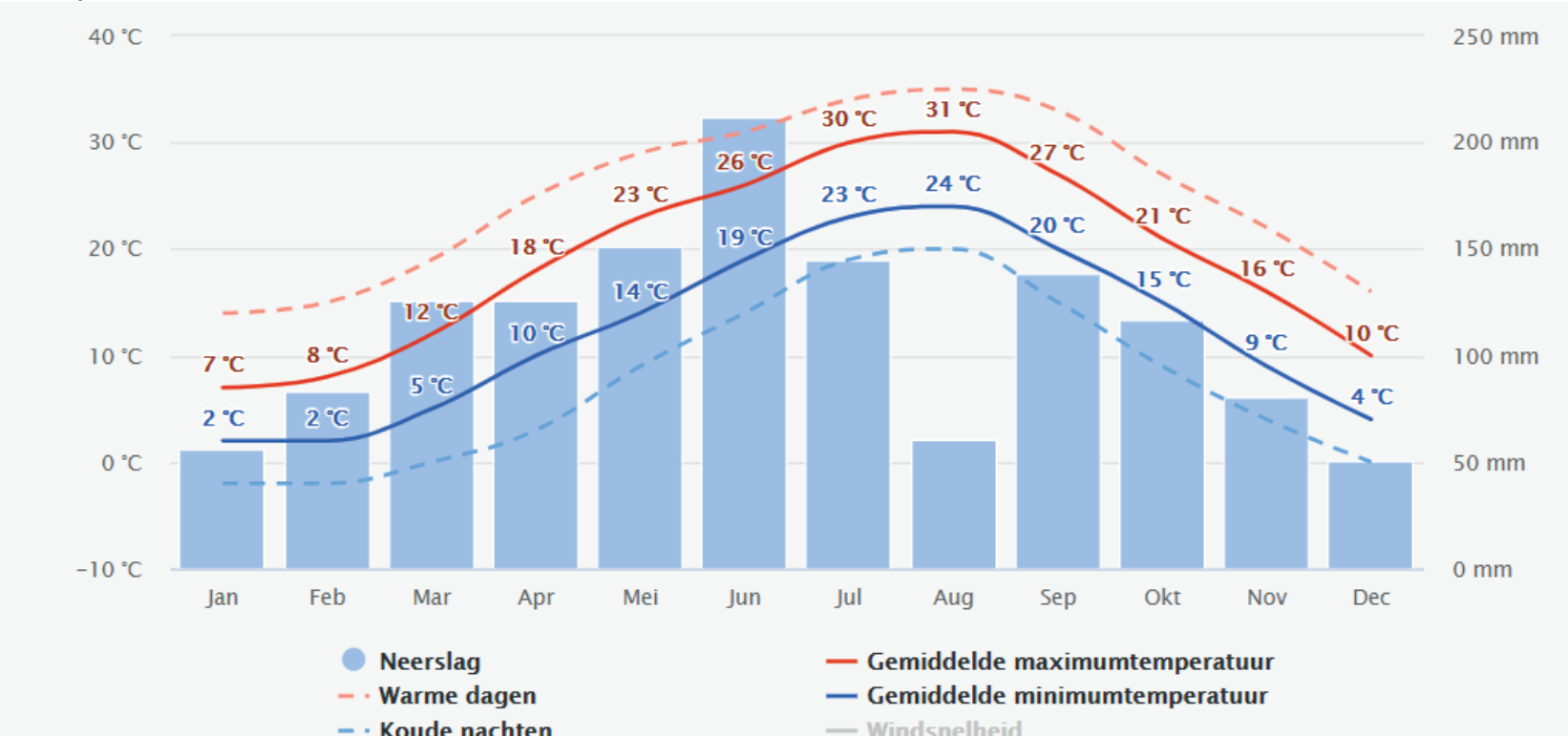
Bron: DiStasio, C. (2015, 9 maart). 3D-printed "Cool Brick" cools a room using only water. Inhabitat - Green Design, Innovation, Architecture, Green Building. Geraadpleegd op 28 oktober 2021, van <https://inhabitat.com/3d-printed-cool-brick-cools-a-room-using-only-water/>

The climate - Osaka

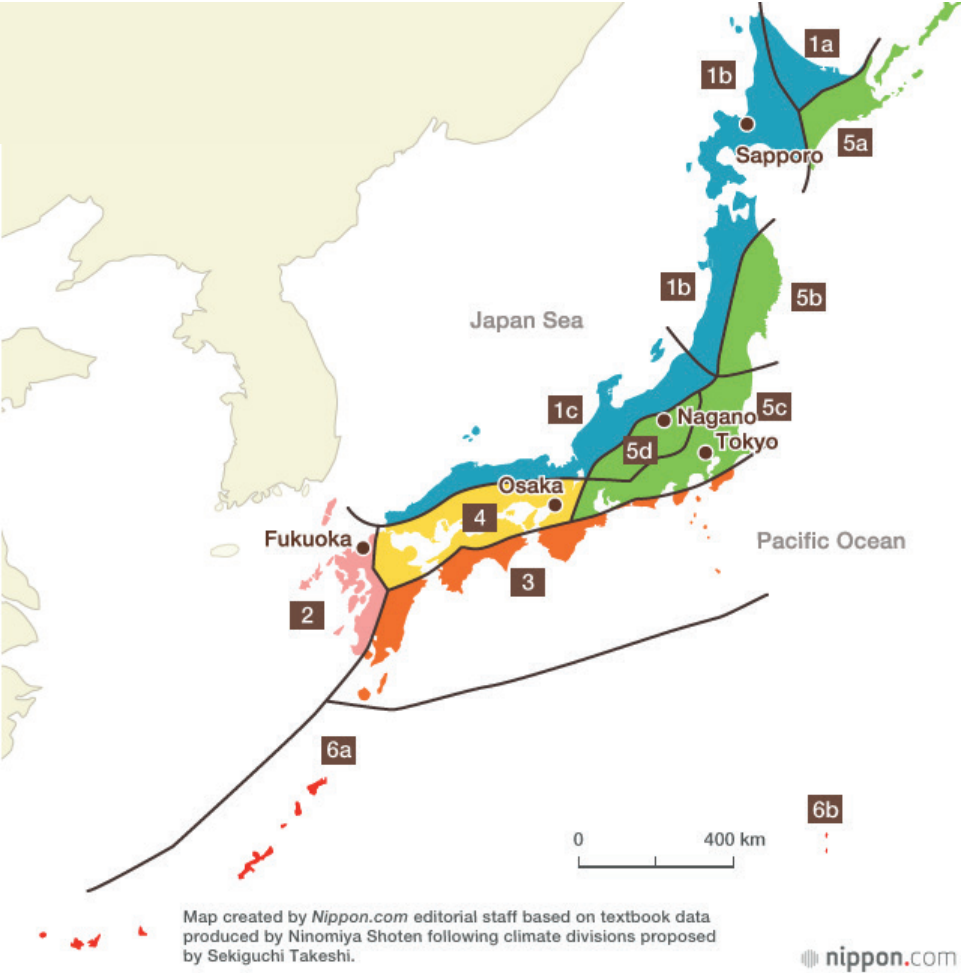
Wind



Temperature and rainfall

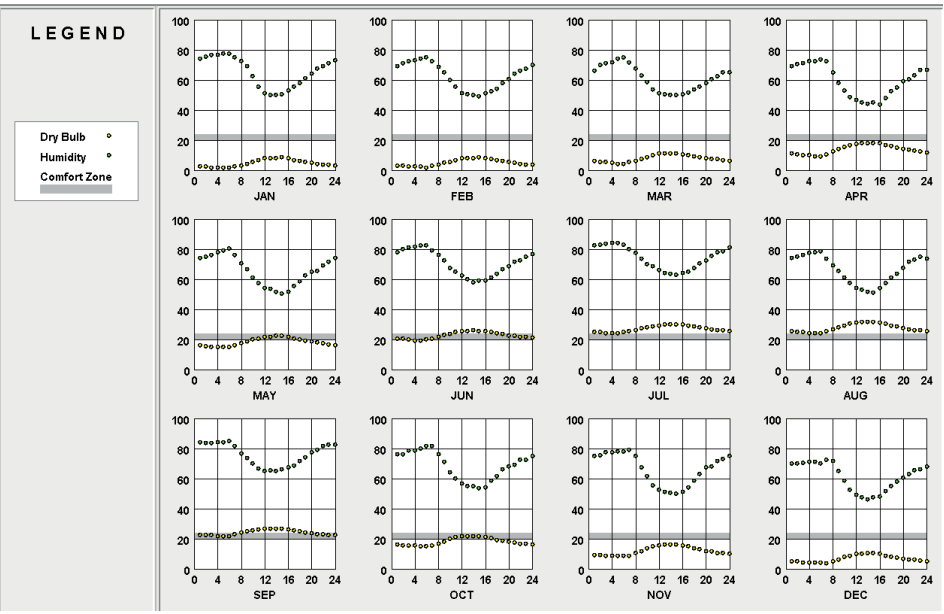


Climate zones



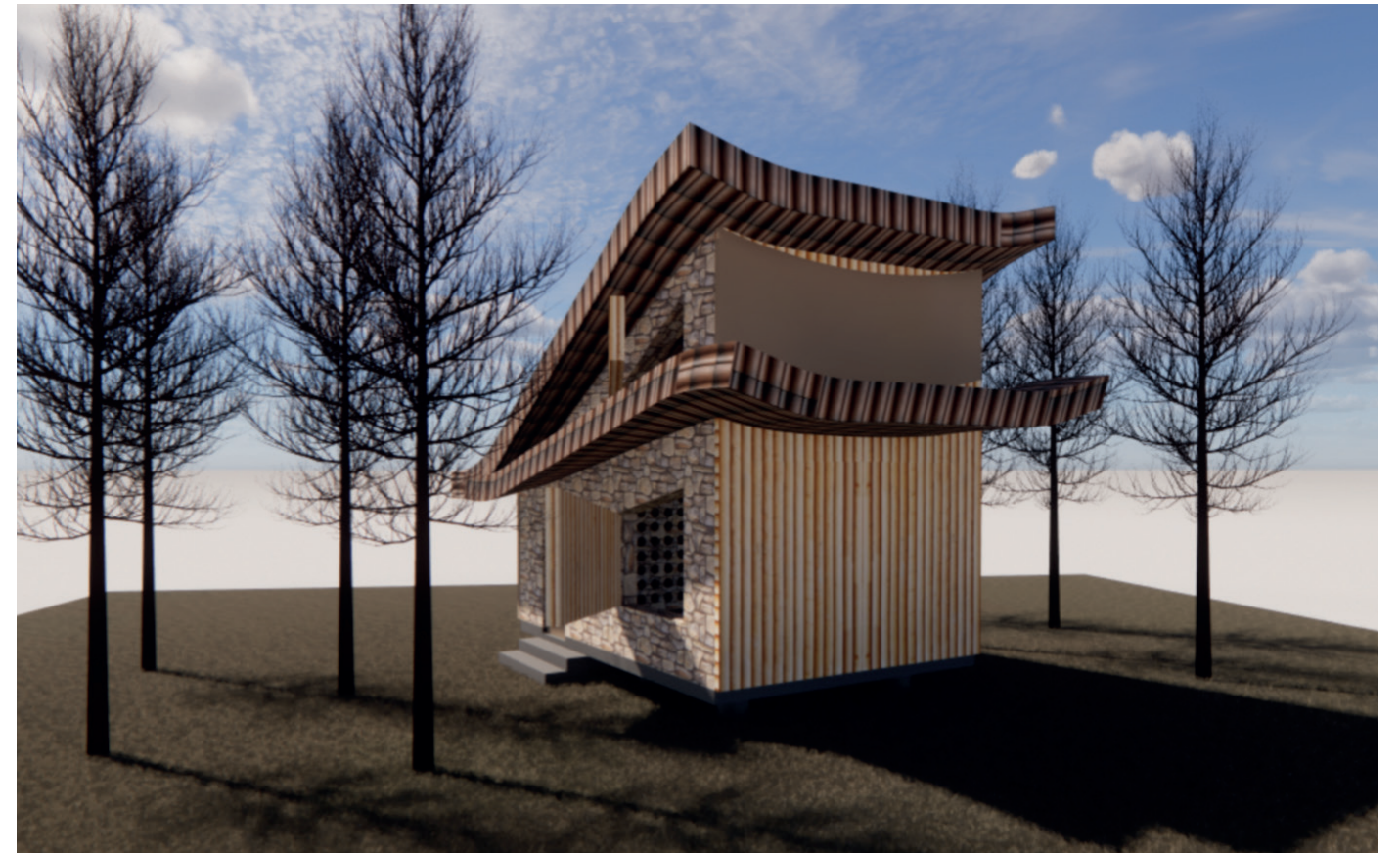
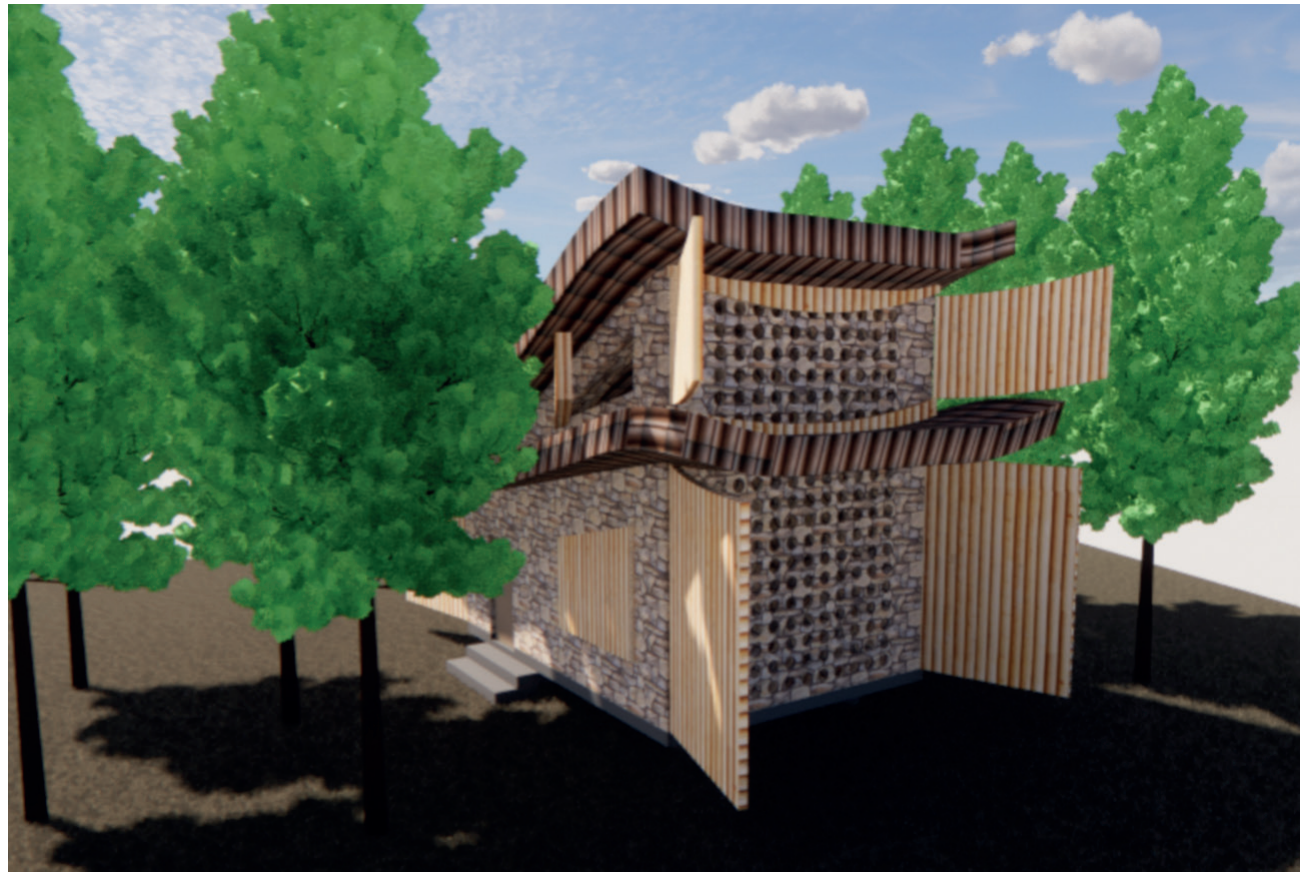
1	Japan Sea	1a Okhotsk	Average temperature below 0°C in winter. Low annual rainfall.
		1b Tōhoku; Hokkaidō	Heavy rainfall in September and winter. Average temperature below 0°C in January and February
		1c Hokuriku; San'in	Average monthly temperature does not fall below 0°C. Heavy snow in winter.
2	2 Kyūshū		Heavy rainfall in June and July.
3	3 Nankai		Heavy rainfall all year round, particularly around June and September.
4	4 Setouchi		Warm with low rainfall all year round
5	East Japan	5a East Hokkaidō	Severely cold in winter. Low rainfall, except around September when it is high.
		5b Sanriku; Jōban	Lower temperatures and less snow compared with the Japan Sea side at this latitude.
		5c Tōkai, Kantō	Heavy rain around June and September to October. Fine weather in winter.
		5d Chūō kōgen	Great seasonal variation in temperature. Severely cold in winter with little accumulation of snow.
6		6a Nansei shotō	Warm with little seasonal variation in temperature. Low rainfall.
		6b Chichijima	Warm with little seasonal variation in temperature. Heavy rainfall in May and November.

Humidity



- Summary:
- High humidity
 - Cold winter / hot summer
 - Wind mainly from the northwest

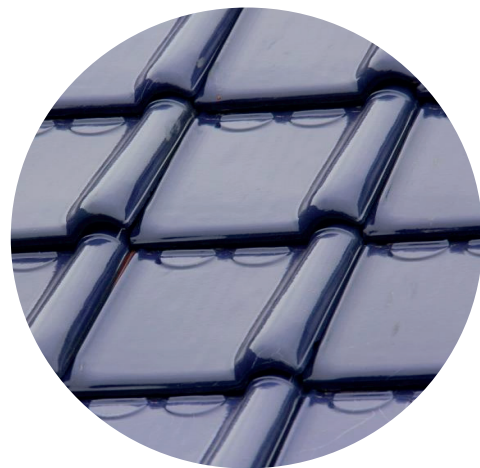
Impressions winter and summer



Climate components



Sloped roof gives opportunity to collect water for the ventallation and or drinkwater/ graywater



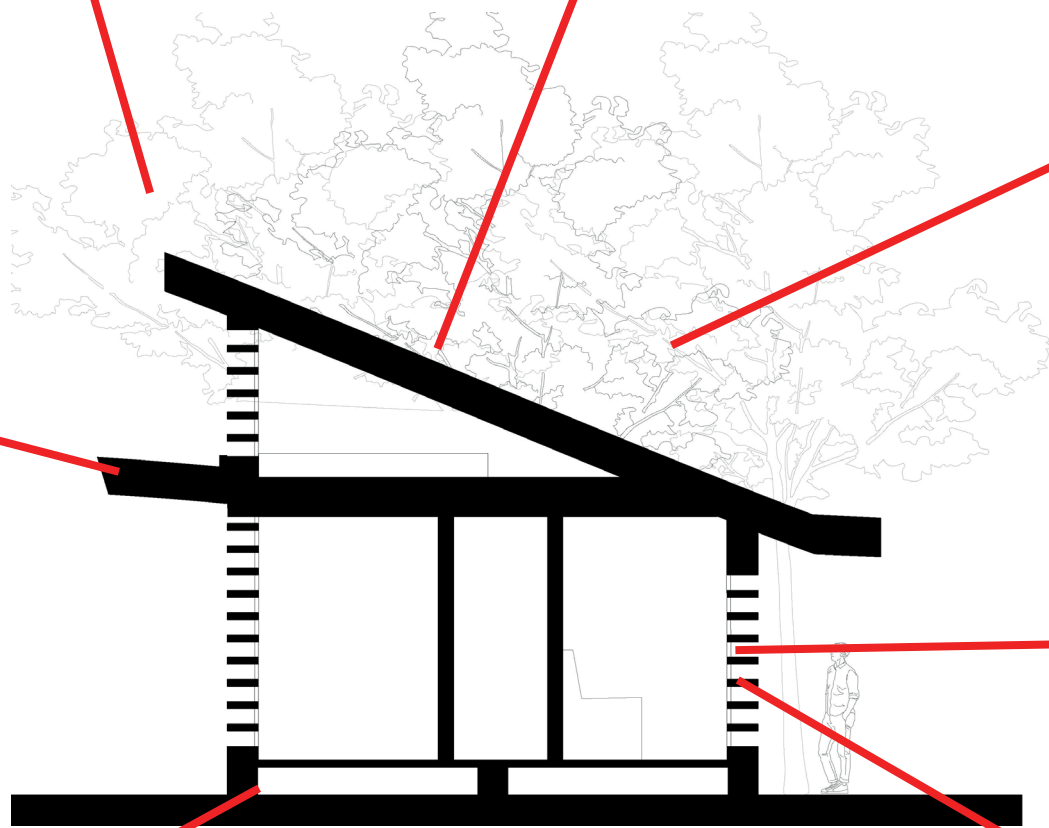
Glazed rooftiles to reflect sun radiation/ solar roof to collect energy as well



Summer green trees to keep the humidity at bay and provide shade in summer
- Native species



Dubble and overhanging roof to provide shade in summer



Filters in the holes to keep the humidity from rising and or bugs to enter the house



House lifted from the ground against floodings, irregular terrain and for ventillation



Air ventilation in summer / isolated panels in the winter and use of local raw materials